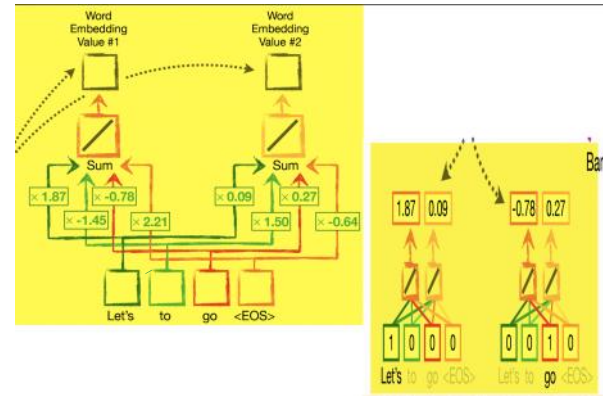


Encoder① Embedding values:

$$E_{\text{Let's}} = [1.87 \ 0.09]$$

$$E_{\text{go}} = [-0.78 \ 0.27]$$

You can find it by taking one-hot vector over the vocabulary =

$$\begin{bmatrix} \text{Let's} \\ \text{to} \\ \text{go} \\ \text{<EOS>} \end{bmatrix}$$
② Positional Encoding

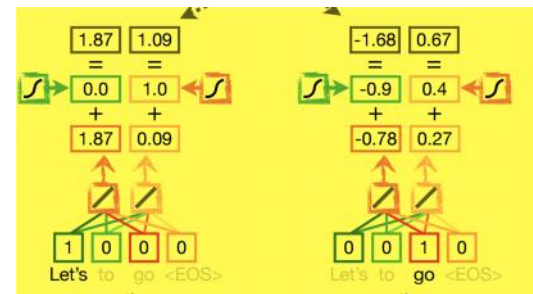
$d_{\text{model}} = 2$ [there are two embedding value for each word]

For Let's

$$E_{\text{Let's}} = [1.87 \ 0.09]$$

$\text{pos} = 0$ [first word in the sentence, $\text{pos} = 0$]

$i = 0, 0$ [$d_{\text{model}} = 2$, for embedding-1 $i = 0$
for embedding-2 $i = 0$]



$$\begin{aligned} PE(\text{pos}=0, 2 \times i=0) &= \sin\left(\frac{\text{pos}}{10000^{\frac{2i}{d_{\text{model}}}}}\right) \\ \text{[for } 1.87] &= \sin\left(\frac{0}{10000^{\frac{2 \times 0}{2}}}\right) = 0 \text{ [in Radian]} \end{aligned}$$

$$\begin{aligned} PE(\text{pos}=0, 2 \times i+1=1) &= \cos\left(\frac{\text{pos}}{10000^{\frac{2i}{d_{\text{model}}}}}\right) \\ \text{[for } 0.09] &= \cos\left(\frac{0}{10000^{\frac{2 \times 0}{2}}}\right) = 1 \text{ [in Radian]} \end{aligned}$$

Note:

If $d_{\text{model}} = 4$, $i = 0, 0, 1, 1$
for embedding-1, $i = 0$
for embedding-2, $i = 0$
for embedding-3, $i = 1$
for embedding-4, $i = 1$

Embedding of Let's after adding P.E.

$$E_{\text{Let's}} = [1.87 \ 0.09] + [0 \ 1] = [1.87, 1.09]$$

For go

$$E_{go} = [-0.78, 0.27]$$

pos = 1 [2nd word in the sentence, pos=1]

$$i = 0, 0$$

$$PE(pos=1, 2 \times i=0) = \sin\left(\frac{pos}{10000^{\frac{2i}{\text{index}}}}\right)$$

[for -0.78]

$$= \sin\left(\frac{1}{10000^{\frac{2 \times 0}{2}}}\right) = 0.84 \text{ [in Radian]}$$

$$PE(pos=1, 2 \times i+1=1) = \cos\left(\frac{pos}{10000^{\frac{2i}{\text{index}}}}\right)$$

[for 0.27]

$$= \cos\left(\frac{1}{10000^{\frac{2 \times 0}{2}}}\right) = 0.54 \text{ [in Radian]}$$

Embedding of go after adding P.E.

$$E_{go} = [-0.78 \ 0.27] + [0.84 \ 0.54] = [0.06 \ 0.81]$$

[The slide uses toy value -0.9 and 0.4]

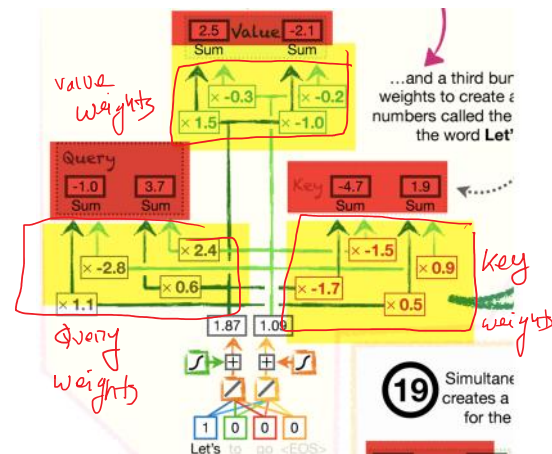
③ Calculate self attention:

The Query, key and value weights

$$Q = \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix} \quad K = \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{bmatrix} \quad V = \begin{bmatrix} 1.5 & -1 \\ -0.3 & -0.2 \end{bmatrix}$$

Q, K, V of Let's

$$E_{Let's} = [1.87 \ 1.09]$$



$$E_{\text{Let's}} = \begin{bmatrix} 1.87 & 1.09 \end{bmatrix}$$

$$Q_{\text{Let's}} = E_{\text{Let's}} \times Q = \begin{bmatrix} 1.87 & 1.09 \end{bmatrix} \times \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 3.7 \end{bmatrix} \quad [\text{You can take approximate values}]$$

$$K_{\text{Let's}} = E_{\text{Let's}} \times K = \begin{bmatrix} 1.87 & 1.09 \end{bmatrix} \times \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.2 \end{bmatrix}$$

$$= \begin{bmatrix} -4.7 & 1.9 \end{bmatrix}$$

$$V_{\text{Let's}} = E_{\text{Let's}} \times V = \begin{bmatrix} 1.87 & 1.09 \end{bmatrix} \times \begin{bmatrix} 1.5 & -1 \\ -0.3 & -0.2 \end{bmatrix}$$

$$= \begin{bmatrix} 2.5 & -2.1 \end{bmatrix}$$

Q, K, V of go

$$E_{\text{go}} = \begin{bmatrix} -1.68 & 0.67 \end{bmatrix} \quad [\text{Let us use the toy values to match the result.}]$$

$$Q_{\text{go}} = E_{\text{go}} \times Q = \begin{bmatrix} -1.68 & 0.67 \end{bmatrix} \times \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix}$$

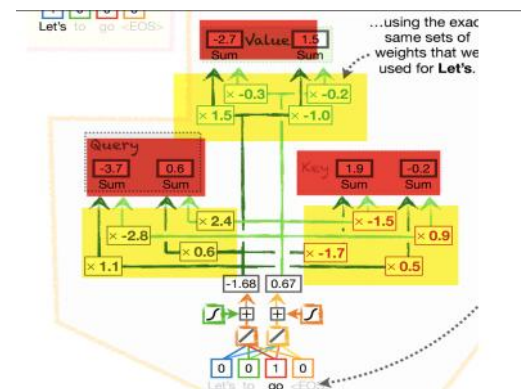
$$= \begin{bmatrix} -3.7 & 0.6 \end{bmatrix}$$

$$K_{\text{go}} = E_{\text{go}} \times K = \begin{bmatrix} -1.68 & 0.67 \end{bmatrix} \times \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.2 \end{bmatrix}$$

$$= \begin{bmatrix} 1.9 & -0.2 \end{bmatrix}$$

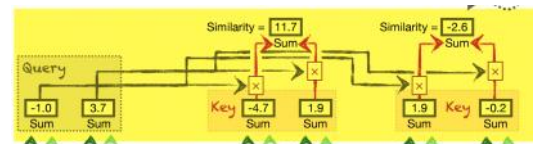
$$V_{\text{go}} = E_{\text{go}} \times V = \begin{bmatrix} -1.68 & 0.67 \end{bmatrix} \times \begin{bmatrix} 1.5 & -1 \\ -0.3 & -0.2 \end{bmatrix}$$

$$= \begin{bmatrix} -2.7 & 1.5 \end{bmatrix}$$



Attention of Let's

- ② We will multiply Query of Let's with Key of go and Key of go



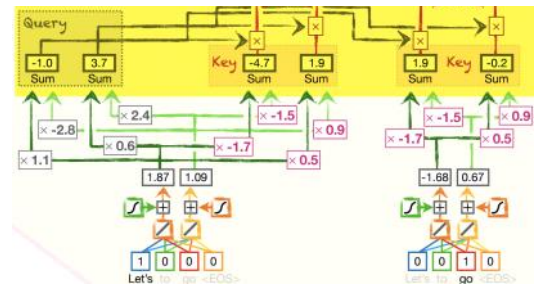
② We will multiply Query of Let's with Key of go and Key of go

$$Q_{\text{Let's}} \cdot K_{\text{Let's}} = \begin{bmatrix} -1 & 3.7 \end{bmatrix} \cdot \begin{bmatrix} -4.7 & 1.9 \end{bmatrix}$$

$$= \{(-1) \times (-4.7)\} + \{3.7 \times 1.9\} = 11.7$$

$$Q_{\text{Let's}} \cdot K_{\text{go}} = \begin{bmatrix} -1 & 3.7 \end{bmatrix} \cdot \begin{bmatrix} 1.9 & -0.2 \end{bmatrix}$$

$$= \{(-1) \times 1.9\} + \{3.7 \times (-0.2)\} = -2.6$$



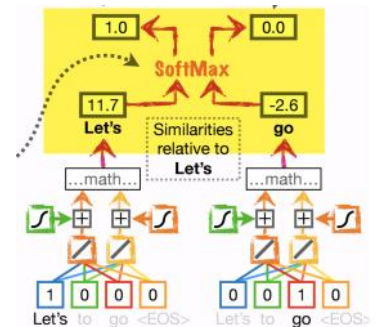
② Do Softmax to find Similarity to Let's

$$\text{Softmax}(Q_{\text{Let's}} \cdot K_{\text{Let's}}, Q_{\text{Let's}} \cdot K_{\text{go}})$$

$$= \text{softmax}(11.7, -2.6)$$

$$= \left[\frac{e^{11.7}}{e^{11.7} + e^{-2.6}}, \frac{e^{-2.6}}{e^{11.7} + e^{-2.6}} \right]$$

$$= [1, 0]$$

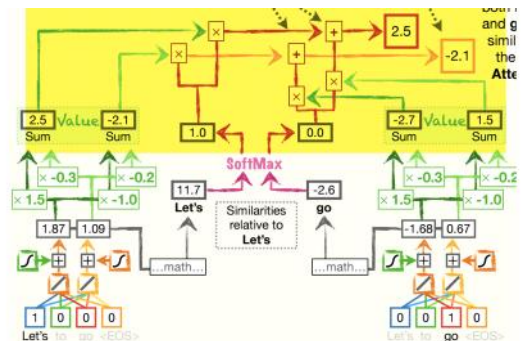


③ Attention for Let's

$$(V_{\text{Let's}} \times 1) + (V_{\text{go}} \times 0)$$

$$= \begin{bmatrix} 2.5 & -2.1 \end{bmatrix} \times 1 + \begin{bmatrix} -2.7 & 1.5 \end{bmatrix} \times 0$$

$$= \begin{bmatrix} 2.5 & -2.1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \end{bmatrix} = \begin{bmatrix} 2.5 & -2.1 \end{bmatrix}$$

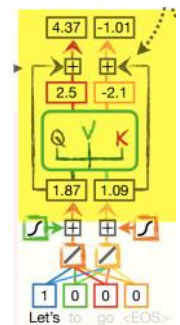


④ Residual connection [update Embedding of Let's]

$$E_{\text{Let's}} + \text{Attention of Let's}$$

$$= \begin{bmatrix} 1.87 & 2.09 \end{bmatrix} + \begin{bmatrix} 2.5 & -2.1 \end{bmatrix}$$

$$= \begin{bmatrix} 4.37 & -1.01 \end{bmatrix}$$



Attention of go

... with Key of Let's

Attention of go

② We will multiply Query of go with Key of Let's and Key of go

$$Q_{go} \cdot K_{Let's} = [-3.7 \ 0.6] \cdot [-4.7 \ 1.9] \\ = \{(-3.7) \times (-4.7)\} + \{0.6 \times 1.9\} = 18.5$$

$$Q_{go} \cdot K_{go} = [-3.7 \ 0.6] \cdot [1.9 \ -0.2] \\ = \{(-3.7) \times 1.9\} + \{0.6 \times (-0.2)\} = -7.2$$

③ Do softmax to find similarity to go

$$\text{Softmax}(Q_{go} \cdot K_{Let's}, Q_{go} \cdot K_{go}) \\ = \text{softmax}(18.5, -7.2) \\ = \left[\frac{e^{18.5}}{e^{18.5} + e^{-7.2}}, \frac{e^{-7.2}}{e^{18.5} + e^{-7.2}} \right] \\ = [1, 0]$$

③ Attention for go

$$(V_{Let's} \times 1) + (V_{go} \times 0) \\ = [2.5 \ -2.1] \times 1 + [-2.7 \ 1.5] \times 0 \\ = [2.5 \ -2.1]$$

④ Residual connection [update Embedding of go]

$$E_{go} + \text{Attention of go} \\ = [-1.68 \ 0.67] + [2.5 \ -2.1] \\ = [0.82 \ -1.43]$$

